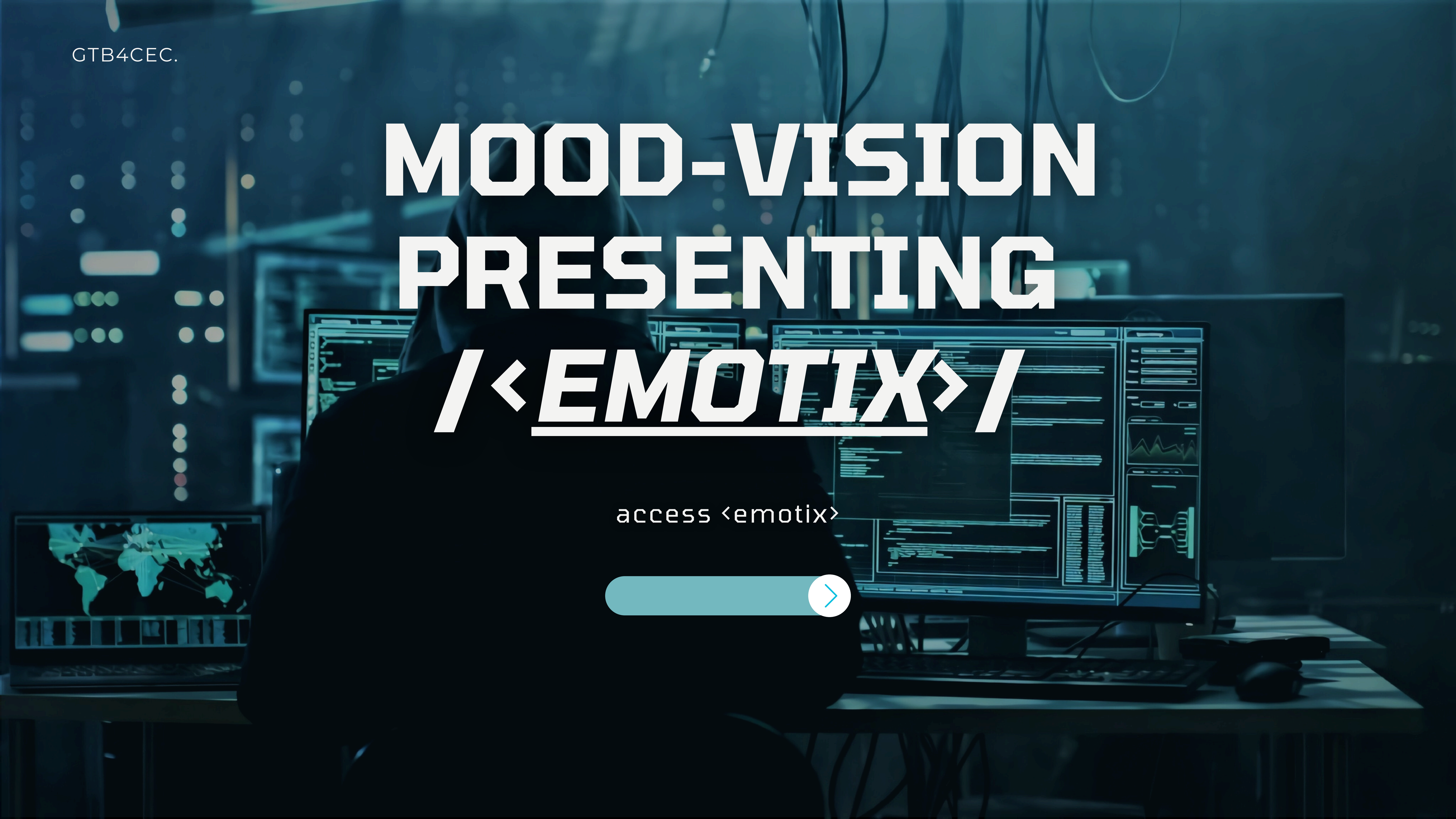

GTB4CEC.

MOOD-VISION PRESENTING

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TEAM MEMBERS

TANISHQ MADHWANI [TEAM LEADER]

PARTH VATS

LAVANYA JAIN

YUG SHARMA

OVERVIEW:*FER*

A Deep Learning Approach using Python and CNN

01 **Why FER** :
To develop a Real-Time System that detects and classifies human emotions from image feeded.

02 **Core Technology:**
Deep Learning: Utilizes Convolutional Neural Networks (CNN)[model : yolo V11], the leading algorithm in Computer Vision, for high-accuracy feature extraction

03 **Classification Output:**
The model classifies facial expressions into 7 Fundamental Emotional States:

->Happy ->Neutral
->Sad ->Surprise
->Angry ->Disgust
->Fear

BARRIERS IN EXPRESSING EMOTIONS

Technical limitation in recognition

Difficulty in accurately interpreting subtlety, mixed emotions, sarcasm, and context in human expression (text, voice, face).

Conceptual gaps in expression

AI systems lack genuine emotional intelligence (empathy, self-awareness, personal experience) and only simulate or categorize emotions based on patterns, leading to outputs that can feel insensitive or hollow.

WORKING APPROACH

01 As per our approach/model, we are concerned with the classification of the emotions from the provided facial expressions in the form of digital image i.e. (INPUT)

02 To develop such a model, we need an algorithm that can work efficiently with the provided dataset to correctly detect the emotions of the facial expression given to it as an input in the form of digital image.

03 Hence, to implement this model we have used different algorithms & a dataset to implement working model.

TECHNICAL STACK : PYTHON LIBRARIES

TENSORFLOW

Primary deep learning framework

PANDAS,NUMPY

Data Handling & Numerical Operations

OPENCV[CV2]

Image & Vision processing

SCIKIT LEARNER

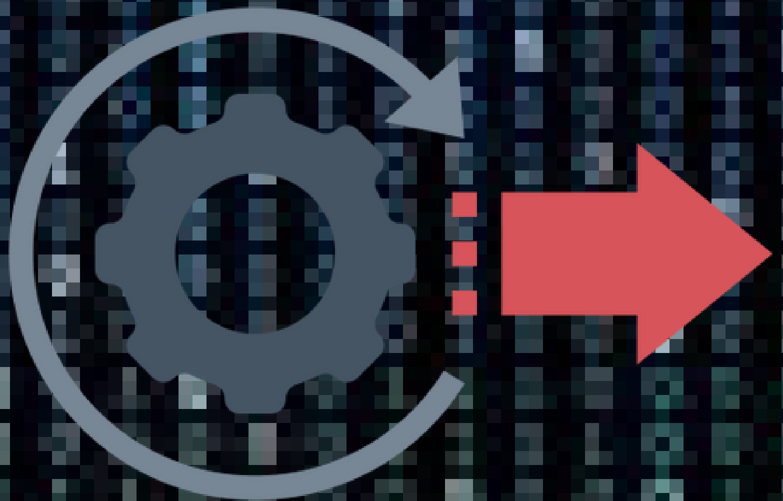
Provides the LabelEncoder to convert human-readable emotion labels (e.g., 'Happy') into the numerical classes required for model training.

<EMOTIX> WORKFLOW

INPUT IMAGE

ANALYSING THE INPUT IMG.

PROCESSING & CHECKING FROM THE GIVEN DATASET



FACIAL EMOTION

Image Emotion Detector: Inference Loop

Face Detection

Tools: OpenCV [yolo V11]

Action: Locates the human face image.

Prediction

Tools: Trained CNN Model [yolo V11]

Action: The preprocessed face is passed to the model, which outputs a probability distribution across the 7 emotion classes

Processing

Tools: Cropping & Resizing

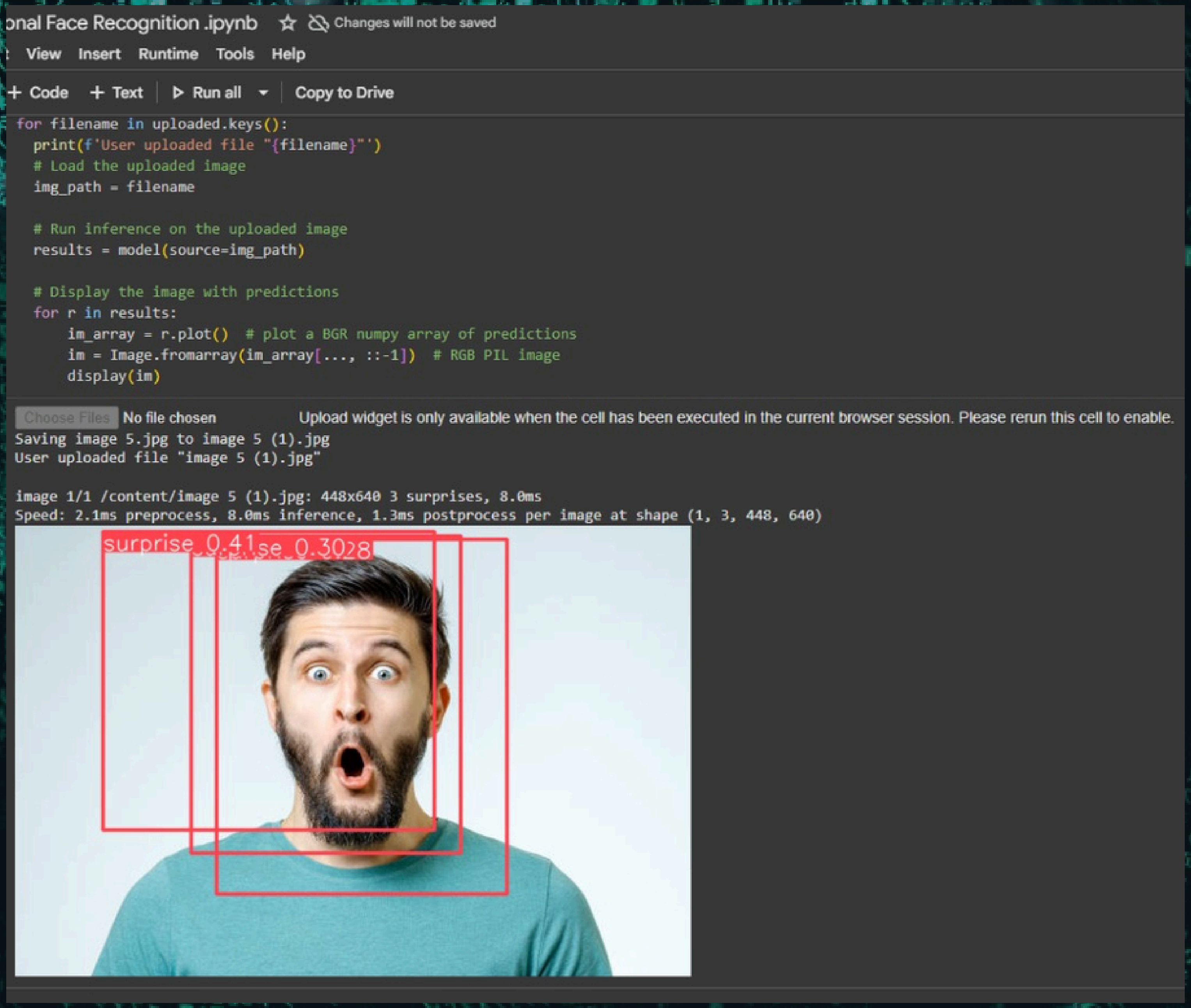
Action: The detected face region is cropped, resized to the mandatory (48×48) pixels, and converted to grayscale.

Output Display

Tools: Display Logic

Action: The emotion with the highest probability is selected and displayed as a text label overlay on the face's bounding box.

OUTPUT INTERFACE



DATA PREPERATION

Image Standard Size

IAction: 48x48
Impact: Ensures efficiency and reduces computational load for real-time processing and training.

Normalization

Action: Scale to [0, 1]
Impact: Pixel values (0-255) are divided by 255. This standardizes the input and improves the stability and speed of model training.

DATA PREPERATION FOR CNN.
FROM IMAGE TO ARAAY

Color Conversion

Action: Grayscale
Impact: Reduces the complexity of the data by converting from 3 color channels (RGB) to just 1 channel.

Label Conversion

Action: One-Hot Encoding
Impact: Transforms string labels (e.g., "Happy") into numerical vectors required for the model's multi-class output layer.

Future scope

Transfer Learning

Utilizing pre-trained, complex models (like VGG or ResNet) trained on millions of general images.

Reuses sophisticated visual knowledge, providing an instant high-level feature base for FER.

Fine-Tuning

We will only train the final layers of the pre-trained model on our 7-emotion dataset.

Significantly reduces training time and computational cost, leading to faster development.